

Multiple Choice

1. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: C - True, False

2. **Answer:** A

Options: A) the open first quadrant; B) a closed curve; C) a ray in the open first quadrant; D) a single point

Note: Correct option: A - the open first quadrant

3. **Answer:** B

Options: A) 0; B) 4; C) 2; D) 6

Note: Correct option: B - 4

4. **Answer:** C

Options: A) 1; B) 2; C) 3; D) 4

Note: Correct option: C - 3

5. **Answer:** A

Options: A) Every compact space is complete; B) Every complete space is compact; C) Neither (a) nor (b).; D) Both (a) and (b).

Note: Correct option: A - Every compact space is complete

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '45'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** We notice that the denominator on the left factors, giving us

$$\frac{2x + 4}{(x - 1)(x + 5)} = \frac{2 - x}{x - 1}.$$

As long as $x \neq 1$ we are allowed to cancel $x - 1$ from the denominators, giving

$$\frac{2x + 4}{x + 5} = 2 - x.$$

Now we can cross-multiply to find

$$2x + 4 = (2 - x)(x + 5) = -x^2 - 3x + 10.$$

We simplify this to

$$x^2 + 5x - 6 = 0$$

and then factor to

$$(x - 1)(x + 6) = 0.$$

Notice that since $x - 1$ is in the denominator of the original equation, $x = 1$ is an extraneous solution. However $x = -6$ does solve the original equation.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** True, True. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key